

# Graph Data Analysis

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## 1 Introduction

This analysis uses the USAir97 dataset, a network representing airline connections in the United States. The dataset features nodes as airports and edges as flight routes between them. By studying the structure of this network such as degree distribution, clustering coefficients, distance, diameter, average distance and comparing it to a random graph model, the analysis sheds light on key differences in community structure, centrality metrics, and connectivity patterns. The study emphasizes that the USAir97 network exhibits non-random characteristics, with features such as:

- Greater variance in community size.
- Higher clustering coefficients indicating dense, localized clusters.
- Variability in connectivity reflecting the unique hub-and-spoke structure of airline networks.

## 2 Key Insights and Comparisons

### 2.1 Network Structure

#### Original Graph:

- Regarding degree distribution in the USAir97, there are really few nodes that have a really high degree and a majority of nodes have small degrees.
- The clustering coefficient are widely distributed between 0 and 1 which shows that the graph nodes are tightly connected.
- the degree correlation of this graph is negative, which let us think that high degree nodes tend to connect to low degree nodes (which is logic when you think about a big state airport in the US that serves small states airports)

**Random Graph:**

- Degree distribution in random graphs is much more heterogenous, as all nodes are most likely not big differences in degree, which is most of the case in random graphs.
- The clustering coefficient is concentrated between 0 and 0.1 with a few nodes with more than 0.1 which means that the nodes are loosely connected in this graph.

## 2.2 Community Statistics

**Original Graph:**

- Communities in the USAir97 network exhibit varying sizes and degrees of connectivity.
- Higher clustering coefficients suggest dense, localized connections within communities.

**Random Graph:**

- Communities are more uniform in size and have homogeneous connectivity.
- The clustering coefficients are significantly lower, reflecting the random placement of the edges.

## 2.3 Centrality Measures

**Degree Centrality:**

- Airports with high degrees correspond to major hubs for instance; Atlanta or Chicago.

**Closeness Centrality:**

- High closeness centrality nodes are typically well-connected hubs, reducing average shortest path distances.

**Betweenness Centrality:**

- Nodes with high betweenness often act as critical bridges between communities, ensuring the flow of traffic across the network.

## 2.4 Correlations Between Centrality Measures

The analysis revealed correlations between centrality measures, which were further explained with the help of AI.

#### **Degree Centrality and Closeness Centrality:**

- A strong positive correlation indicates that highly connected nodes tend to shorten path lengths across the network.

#### **Degree Centrality and Between-ness Centrality:**

- Although generally correlated, exceptions occur in sparse or modular networks.

### **2.5 Other Observations**

Certain airports with moderate closeness centrality exhibited unexpectedly high betweenness. This reflects their role as critical connectors between otherwise distant regions.

## **3 Role of AI Assistance**

AI assistance via **Gemini** was instrumental in deepening our understanding. It Clarified theoretical concepts and validated our observations. Key prompts included:

- "Why closeness centrality and degree centrality seem to be correlated?"
- "What explains the positive correlation between degree and betweenness centralities in most graphs?"
- "Why do some nodes with moderate closeness centrality show high betweenness?"
- "Is the correlation between degree centrality and betweenness centrality a general property of graphs?"

The insights provided by **Gemini** enriched the interpretation of complex patterns and ensured accurate conclusions.

## **4 Conclusion**

This analysis underscores the value of studying real-world network structures like the USAir97 graph. By leveraging AI tools, the study effectively demonstrated key differences between real and random graphs, providing insights into the connectivity, centrality, and community structures that define airline networks. It shows that the real graph is well structured with logical connection, while the random graph shows a lack of structure. These findings can guide future network modeling and optimization efforts.